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The Perception of Perceived Choice

Implications for safety, health, and life satisfaction

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The freedom to choose is invaluable for human well-being and life satisfaction. Yet, we often take this crucial human dynamic for granted, without realizing the dramatic influence of perceived choice. Consider, for example, the extent to which the pleasure of a weekend break from a weekly work routine is determined by the opportunity to choose what to do each day and when to do it by starting from choosing when to awake in the morning to deciding your daily activities, some of which might require more energy and effort than that required to accomplish typical work assignments.

In other words, your greater enjoyment performing one task over another is often due to the perception of having more personal choice for that task than the other.

Given that human welfare or subjective wellbeing (SWB) is enhanced by the perception of personal choice, people need to know how to increase perceptions of choice within themselves and among others—from young children and teenagers to family, friends, and work colleagues.

This paper explains how to make that happen for occupational safety and health (OSH). While personal experience and commonsense teach us the positive impact of perceived choice, psychological science has demonstrated the power of perceived choice in OSH-related situations..

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The power of perceived choice might not be appreciated while it is experienced, but personal choice is surely recognized when it is threatened, as illustrated humorously in Figure 1, and observed frequently when COVID facemasks were mandated. Some individuals perceived that facemask requirement as a threat to their personal freedom, and they actively resisted requests to comply with the mandate (e.g., on an airplane). Social psychologists refer to such attempts to maintain personal choice as "psychological reactance" (Brehm, 1966), and behavioral scientists call such contrary behavior counter control" (Skinner, 1971). Regardless of the label for behavioral resistance to a mandate, reacting to maintain personal choice can be extremely counterproductive for OSH. For instance, the author once met an employee who wore safety frames—not safety glasses, only safety frames—because he had removed the lenses. Whenever that employee's supervisor passed by, this employee would look at him and wave. The coworkers of this individual actually applauded him for his noncompliance. Why, because at that facility most employees perceived safety as a top-down mandate that restricted individual freedom or perceived choice.



Figure 1: Mandates can be counter productive

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In fact, this particular employee increased his social status among his work colleagues by demonstrating counter control or psychological reactance.

Self-Directed Behavior

At times, people need an external accountability system to keep them motivated. Psychologists call these “extrinsic motivators,” as when managers and teachers use paychecks and grades to keep employees and students on track, respectively. Sometimes people develop self-motivation within the context of an external motivator. In other words, it is possible to establish conditions that facilitate self-motivation and self-directed behavior. In fact, the perception of choice or “autonomy” is a primary determinant of self-motivated behavior (Deci & Flaste, 1995).

Imagine helping a young boy dress in the nice clothes you have selected for him to wear on his first day of school. This could feel like top-down control to the child, and he might resist in order to assert his personal freedom or individuality. Now consider an alternative approach: Select two school outfits you find equally appropriate for the boy to wear, and let him choose between the two. Now the young boy will be less resistant because he perceives some choice in the clothes he will wear to school.

Switching to a workplace scenario, consider two approaches to soliciting problem-solving assistance from a work team. A common approach is for the supervisor to define both the problem and the solution, and then to assign certain employees relevant problem-solving tasks. Alternatively, the supervisor might solicit volunteers to perform the various problem-solving tasks, and thereby integrate some perceived choice into the scenario.

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A more effective approach would be as follows: Describe the problem to the work team, but ask the workers for their solution possibilities, including the various problem-solving tasks required. Obviously, this approach will require much more interpersonal dialogue, and therefore it will be substantially less efficient than the first approach. However, the work team might derive a more effective solution. Even if the problem-solving tasks derived by the employees are the same as those proposed by the supervisor using the first approach, the perceived choice and self-motivation inspired by the second "less-efficient" approach would be well worth the extra time and effort.

Participative Management

Many readers are familiar with the term "participative management," which essentially means employees are provided with some personal choice during the planning, execution, and/or the evaluation of their jobs. The result: more self-motivation, engagement, and life satisfaction among the workforce. As indicated above, participative management takes more time and is therefore less efficient than a management-directed approach, but the gain in effectiveness is typically well worth the loss of efficiency.

A Personal Example

More than 45 years ago, the author explored participative management and perceived choice within a university course. I was teaching two sections of social psychology—one at 8:00am on Monday, Wednesday, and Friday, and the other at 11:00am on those same days.

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There were about 75 students in each class. On the first day of the 8:00am class, I did not distribute a typical prepared syllabus with weekly reading assignments. Instead, I distributed only a general outline of the course that introduced the textbook, the course objectives, and the basic criteria for assigning final grades: a quiz on each textbook chapter and a comprehensive final exam on class lectures, discussions, and demonstrations (Geller, 2013).

In an open discussion and voting process, the 8:00am class was given the opportunity to choose the order in which the ten textbook chapters would be assigned for reading, and for classroom discussions and quizzes. The students could submit multiple-choice questions for potential use on the ten chapter quizzes, and they could hand in short-answer and discussion questions for possible inclusion on the final exam. The 11:00am class received the order of textbook chapters selected previously by the 8:00am class, and this class was not given an opportunity to submit quiz or exam questions.

This resulted in a Choice and an Assigned classroom condition. Two undergraduate research students attended each of these classes, posing as regular students; and they systematically recorded the frequency of student behaviors that reflected class participation. These trained observers did not know about the Choice versus Assigned manipulation.

From the day the students in the 8:00am class voted on the textbook assignments, this class seemed more engaged than the later 11:00am class. Indeed, this perception was verified by the participation records of the two classroom observers. Although several students from the 8:00am class submitted potential quiz and final-exam questions, none of those questions were used exactly as written. Each class received the same quizzes and final exam.

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The ten quiz grades, the final-exam scores, and the teaching-evaluation scores from the standard university forms distributed on the last class day were significantly higher in the Choice class than in the Assigned class.

Several factors could have influenced the dramatic group differences, but it is likely the Choice versus Assigned manipulation was a critical factor. The initial opportunity to choose the order of class reading assignments increased students' motivation and class participation from the start, and this extra motivation and involvement led to more involvement, perceived choice, self-motivation, and personal learning.

The students' attitude toward the early-morning class improved as a result of experiencing more perceived choice, personal control, and engagement. It is likely the Choice condition was especially powerful because the choice opportunities for that 8:00am class were so different than the traditional top- down classroom atmosphere at the time, exemplified by the organization of the 11:00am class. In other words, the contrast of the Choice class with the students' other university classes made the choice opportunities in the 8:00am class especially salient, meaningful, and motivating.

Behavior-Based Safety

Most readers are familiar with the concept of "behavior-based safety" or BBS, which the author and his graduate students introduced in 1979 at several Ford Motor Company facilities (Geller, Lehman, & Kalsher, 1987). This innovative approach to OSH became popular immediately, and was disseminated nationwide and then worldwide by several consulting firms. The popularity and success of BBS was undoubtedly due to its participative management qualities, which was notably different from the traditional, rule-governing approach to OSH.

Rather than merely following the safety regulations declared by the safety director of a company and the governmental Occupational Safety and Health Administration (OSHA), employees practicing BBS develop their own checklists of safe and at-risk behaviors for various work settings. Then they select times to use this critical behavior checklist to observe the work

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behaviors of their coworkers, following their permission (e.g., Geller, 1996, 2001, 2005; Geller & Williams, 2001).

Risk Compensation

Safety professionals/leaders should be aware of a research-supported phenomenon labeled risk compensation (Peltzman, 1975) or risk homeostasis (Wilde, 1982). Specifically, these terms refer to the evidence-based phenomenon that an intervention designed to prevent or reduce unintentional injuries (e.g., personal protective equipment, machine guards, and vehicle safety belts) reduces a person's perception of risk, and thereby influences the performance of at-risk behavior. For example, when Jansson (1994) requested habitual non-users of vehicle safety belts to buckle up, they drove faster, followed more closely behind vehicles in front of them, changed lanes at higher speeds, and braked later when approaching an obstacle.

More recently, Hasanzadeh, de la Garza, and Geller (2020a,b) immersed participants in a mixed-virtual reality scenario that simulated a roofing task. The researchers monitored the workers' behaviors and physiological responses under three levels of safety-related intervention, and they found significant risk compensation. For example, when using fall protection, the participants spent more time exposing themselves to the risk of falling by working closer to and leaning over the edge of the 20-foot-high virtual roof.

Research is needed to determine whether educating people about this risk-compensation phenomenon can reduce its detriment to OSH. On the other hand, behavioral science has demonstrated that perceived choice can not only decrease risk compensation, it can influence a positive contrasting effect: response generalization. Specifically, Ludwig and Geller (1997) evaluated the impact of a participative management approach to increasing a particular safe-driving behavior among pizza-delivery drivers. The targeted safe behavior was delivery drivers stopping their vehicle completely before exiting the store's parking lot to deliver a pizza "in 30 minutes or less."

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Researchers unobtrusively observed three driving behaviors of the pizza delivers when they were leaving a popular Domino's Pizza shop in two towns—Blacksburg and Christiansburg, VA. At the Blacksburg Domino's Pizza, a goal of "75% complete stopping" was derived from an interactive group discussion among the store manager, two researchers, and 26 pizza-delivery drivers (n=26). At the Christiansburg Domino's Pizza, a complete-stopping goal of 75% was assigned by the store manager after he delivered a passionate presentation of a need for safer driving practices to the 24 pizza-delivery drivers and two researchers in attendance.

The behavioral observations evidenced a significant benefit of the participative management approach. The pizza-delivery drivers at both stores increased the percentage of times they stopped completely at the target intersection equivalently, and they met the 75% goal. However, the pizza delivers who participated in determining that complete-stopping goal evidenced a significant increase in the occurrence of two other safe-driving behaviors that had not been mentioned during the group meetings: the use of turn signals and safety belts. Thus, applying participative management to set the complete-stopping goal influenced response generalization, which is the opposite of risk compensation.

More recently, the author and his students (Oliver, Browning, Davis, & Geller, 2021) observed an analogous response-generalization effect among students and faculty on the Virginia Tech campus. Specifically, those individuals who wore a COVID-prevention facemask

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outdoors where facemask-wearing was not mandated maintained a greater interpersonal distance from others than did those who were not masked while walking outdoors. As with the pizza-delivery drivers who chose their safe-driving goal, choosing to perform one safe behavior (i.e., wear a facemask) generalized to the performance of another safe behavior (i.e., keep a safe interpersonal distance), which is contrary to risk compensation.

Activating Cognitive Dissonance

People need to feel congruence or consistency between what they think, what they feel, and what they do. We experience tension or cognitive dissonance (Festinger, 1957) when perceiving an inconsistency between a behavior we choose to perform and another behavior, cognition, or personal value. The key word in that sentence is choose. When we feel coerced or incentivized into doing something, we do not feel uncomfortable if that behavior is inconsistent with a personal attitude, value, or another behavior. Can an OSH intervention activate cognitive dissonance, and thereby increase personal responsibility for injury prevention? That question is explored here, which includes the critical relevance of perceived choice.

Recognizing an inconsistency between an internal conviction and self-directed behavior causes mental tension and self-motivation to restore congruity between a behavior and one's belief, attitude, or value. Simply put, we want our actions to reflect our values, and vice versa. When an inconsistency between a value and a self-directed behavior is experienced, behavior is typically adjusted to match the value, rather than changing a value to match the behavior.

Safety as a Value

Many years ago, I proposed that safety should be considered a "value" rather than a "priority," and I offered a reasonable rationale for this, including the interpretation that priorities change as a function of situational factors, but values do not change so readily (Geller, 2003).

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When I have asked audiences at safety conferences whether they hold safety as a value, almost everyone raises a hand to affirm a collective "Yes." Likewise, when this question is posed to individuals, an assertive confirmation is invariably declared. In fact, some say, "Safety is more than a value to me and my work team, it's a core value." "Okay," I reply, "Safety is a value linked to everyone of your priorities. So regardless of the circumstances, including outside demands on your time, you attempt to be as safe as you can be, right?" Surprisingly, most individuals have answered "Yes" to this idealistic proposal, especially in public settings. Actually, we should not be surprised by this response, since in our culture "safety" goes along with "motherhood, baseball, and apple pie."
From Value Affirmation to OSH Behavior.

An intervention process to activate cognitive dissonance and safety-related behavior is simple and straightforward. Inspire people to declare safety as a value linked to the changing priorities of every workday. Then specify behaviors that are compatible versus incompatible with that value statement. Cognitive dissonance can be activated whenever someone points out a behavior that is inconsistent with safety as a core value.

For example, after observing an at-risk work behavior, remind the performer of the group consensus that safety is considered a value at their workplace. If the individual realizes the inconsistency, that worker should experience an inconsistency and proceed to resolve the cognitive dissonance or behavior-value imbalance by substituting safe behavior for the observed at-risk behavior. However, please note the fundamental influence of perceived choice here. The individual's behavioral conviction should not be viewed as controlled by extrinsic

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contingencies, like incentives, disincentives, or peer pressure, but by a self-motivated decision to demonstrate safety as a core value. In other words, cognitive dissonance is only experienced when the behavior-value inconsistency is perceived as voluntary or self-directed.

More than two decades ago, I described the "hypocrisy effect" as an intervention to motivate the occurrence of safety-related behavior by having participants experience a discrepancy between what they have done in the past and what they should do (Geller, 2000). This intervention process is as follows: 1) Present the rationale for a particular safe behavior; 2) Ask the participants to make a commitment to always choose this safe behavior over designated at-risk practices, and then 3) Ask the participants to list the most recent times they had performed an at-risk alternative to the safe behavior.

Note how this hypocrisy-based intervention activates cognitive dissonance. The objective is to provoke the target individuals into experiencing a discrepancy between their behavior and a commitment or personal-value statement. This triggers cognitive dissonance, which ends when self-directed action restores the imbalance.

The more public the commitment or value affirmation, the greater its impact. Thus, when people attest to safety as a value in the presence of others, they feel a special sense of obligation to live up to their affirmation. Those who hear someone declare safety as a value can readily activate cognitive dissonance within that individual by calling attention to a particular at-risk behavior they observe from that person, which does not reflect safety as a value.

Watch Your Language

The author has indicated previously the negative impact of certain safety-related language on perceptions of choice and self-motivated engagement for OSH (e.g., Geller, 2002, 2020b, 2021). The common phrases: "Safety is a condition of employment," and "All accidents are preventable," stifle a sense of perceived choice or autonomy. In fact, the term "accident" implies a lack of personal control or perceived choice, and can be a commonsense excuse for an injury—"It was just an accident, and it was not anyone's fault." Similarly, when young children have an "accident" in their pants, the presumption is that they could not help it—they had no choice.

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Years ago, the National Highway Traffic Safety Administration (NHTSA) advocated for using the words vehicle "crash" or "collision" instead of traffic "accident." That recommendation has been followed quite consistently by transportation-safety professionals, but not by the media nor the public. Indeed, the media commonly reports about a traffic "accident," rather than a vehicle "crash." Many safety professionals use the term "incident" rather than "accident," but the term "incident" is also used to refer to intentional tragic events like hate crimes and mass shootings.

Most workplace "accidents" are unintentional, but they are certainly not chance occurrences. Specific controllable environmental and/or human factors cause injuries, and most of those factors can be changed to prevent future occurrences of the mishap by conducting a comprehensive injury or close-call analysis. Note the word "analysis" rather than "investigation." What about that term "investigation," as in "accident investigation" or "criminal investigation." Doesn't the word "investigation" imply a search for one particular factor or person to blame for a particular injury? The mindset is "fault-finding" to find one "root cause," rather than "fact-finding" to identify environmental and behavioral factors that employers and/or employees can choose to change or improve in order to prevent a workplace injury.

Some other words commonly used by OSH professionals that stifle a perception of choice are: "mandate" (rather than "expectation"), "requirement" (rather than "opportunity"), "compliance" (rather than "accomplishment"), and "peer pressure" (rather than "peer support"). Such choice-inhibiting language is used habitually or "accidentally," and calls for corrective feedback from actively caring observers (Geller, 2020a). For example, a thoughtful and well-written Vantage Point in the January 2023 issue of Professional Safety used the word "accident" seven times on two pages (Wagner, 2023). Imagine the potential positive language impact if the author of that Vantage Point, who is a best-selling author of scholarship published in the New York Times, Forbes, Harvard Business Review, The Globe, and USA Today, had been requested to use a term other than "accident" in his Vantage Point, such as "unintentional injury."

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Conclusion

The research-supported benefits of inspiring and supporting the perception of personal choice that were explicated in this article are not new to most OSH leaders and readers of Professional Safety. As referenced throughout this presentation, the advantages of facilitating perceived choice in order to increase occurrences of OSH-related behavior have been introduced in a variety of books, journal articles, and magazine columns. For example, two prior Professional Safety articles on safety leadership versus management advised leaders to encourage and support perceived choice, and explained why—from inspiring self-motivation (Geller, 2016) to facilitating response generalization (Geller, 2022a). However, none of those prior presentations discussed all five of the OSH benefits of perceived choice that were discussed here, with the ultimate aim to enhance subjective well-being or human happiness. As expressed convincingly by Rodd Wagner (2023), "Happy employees are safer employees" (p. 38).

The distinct value of perceived choice for enhancing self-directed behavior, participative management, and response generalization, and for implementing BBS and a cognitive-dissonance intervention to motivate occurrences of safe behavior is quite obvious—supported by both commonsense and empirical research. However, incorporating perceived choice in a workplace scenario can be "easier said than done," especially for OSH-related responsibilities. Why, because many workplace cultures have approached OSH with a top-down rule-governing mindset. In such workplaces, the notion of involving workers in OSH-related decision-making seems unrealistic and farfetched—from having employees define safe and at-risk behaviors and participate in injury analyses to involving workers in the implementation and evaluation of interventions to improve OSH. In other words, at many workplaces, OSH is the sole responsibility of the "safety department" and involving workers in creating, implementing, and evaluating OSH-related interventions has not been considered.

Thus, incorporating perceived choice in OSH-related situations can be particularly challenging in some work cultures, suggesting a need to cultivate a work culture in which employees: a) feel included with a sense of belongingness and interdependency regarding OSH, b) contribute to continuous OSH improvement as both a teacher and a learner, and c) feel comfortable suggesting work plans, practices, or policies that are contrary to the status quo. Clarke (2020) refers to such a culture as "psychologically safe"—a culture that promotes and supports interpersonal trust, interdependent collaboration, and continuous improvement.

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Geller (2022b) explained six leadership qualities that are vital to cultivating a psychologically safe culture, as derived from the research of Jim Collins and his team (Collins, 2014). Indeed, cultivating a work culture in which perceived choice is integrated into the variety of OSH-related interventions can feel overwhelming. My suggestion: Start small and build by using OSH-related language that supports rather than stifles the perception of choice. For example, it is not an "accident" but an "unintentional injury" that can be prevented by: a) considering the variety of possible contributing environmental and human factors rather than one "root cause," and b) empowering workers to improve those factors that are within their domain of choice and personal control.

Finally, happiness and subjective well-being (SWB) or life satisfaction can be enhanced by becoming more mindful of the numerous choices we have every day when performing ordinary activities. Instead of becoming mindless of habitual routines, become more mindful of the various possible alternatives to each set of potential behaviors, and appreciate the opportunity to willingly select options.

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Building the competence of another generation of Safety Leaders and Safety champions for the Nigerian industry.

Photo showing some MSc students at the Centre for Occupational Health, Safety & Environment (COHSE), University of Port Harcourt, Nigeria

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Occupational Safety vs. Psychological Safety:
Synergistic connections to optimize an organizational culture

E. Scott Geller

Over the years, a number of terms have been introduced and applied to address the psychology of occupational safety and health (OSH). In that regard, the author introduced behavior-based safety (BBS) in 1979; and over the next decade, the BBS process of using worker-derived behavioral checklist for peer-to- peer observation-and-feedback sessions flourished in popularity and success (Geller, 1997; Krause, Hidley, & Hodson, 1996; Sulzer-Azaroff & Austin, 2000).

Subsequently, the need to address more human dynamics than observable behavior inspired the foundation of people-based safety (PBS), which included the observation and feedback process of BBS (Geller, 2005; 2008).

For numerous applications of BBS, the organizational context or culture has prevented optimal injury-prevention outcomes. This led to the concept of safety culture or climate. OSH consultants and trainers introduced methods to cultivate an organizational safety culture—even a Total Safety Culture (Geller, 1994; 2000; 2001)—that could support a successful BBS or PBS process. Consulting firms began marketing and implementing safety-culture surveys to assess whether an organization was “ready for” an effective BBS process. The most beneficial culture surveys pinpoint specific cultural adjustments needed to support an effective BBS. Recently, a new culture-related concept has entered the human-dynamics domain of OSH: “psychological safety.”

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What is Psychological Safety?

"Psychological safety is being able to show and employ one's self without fear of negative consequences to self-image, status or career" (Kahn, 1990, p.708). In psychologically-safe organizations, participants feel accepted and respected, and believe they will not be humiliated or punished when speaking up with questions, concerns, mistakes, or new ideas. Employees who feel psychologically safe are more engaged in their work; they speak up more often and are motivated to improve the performance of their team and organization (Detert & Trevino, 2008; Edmondson, 1999).

Dr. Timothy Clark (2020a) described four stages of psychological safety: inclusion safety, learner safety, contributor safety, and challenger safety. Inclusion safety is the foundation stage when individuals feel accepted in their group or work team. They sense a comfortable person-state of belongingness and interdependency.

In the next stage—learner safety—individuals participate in ongoing learning and teaching. As learners, they openly demonstrate vulnerability by asking questions and requesting help without fear of ridicule or embarrassment.

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They solicit feedback to be the best they can be; and whenever appropriate, they provide feedback to teach others. In other words, team members at this stage of psychological safety cultivate ongoing continuous improvement as both learners and teachers, depending upon situations and circumstances.

When employees or team members give others supportive and/or corrective feedback, they have reached the stage of "contributor safety." With a make-a-difference mindset, these individuals feel empowered to make meaningful contributions to their team and the organization. Participants achieve the ultimate level of psychological safety when they feel comfortable challenging the status quo. Dr. Clark refers to this state as "intellectual bravery"—a "willingness to disagree, dissent, or challenge the status quo in a setting of social risk in which you could be embarrassed, marginalized, or punished in some way" (Clark, 2020b, p. 10).

Note how these four levels of psychological safety suggest the use of various survey items or questions to determine the level of psychological safety perceived by an individual. More specifically, a culture survey could request employees' opinions regarding the following statements by assigning each statement a score of "1" to "5" on a Likert scale, with "1" for "Totally Disagree" and "5" for "Totally Agree." [Please note that these survey items are only suggestions, and would need to be customized for a particular workplace.]

i. I feel a positive sense of belonging within my work team.
ii. I feel a positive sense of belonging within my work team.
iii. I am comfortable asking my supervisor questions about my job.
iv. I willingly ask my coworkers for their assistance whenever needed.
v. I often provide work-improvement advice to other employees.
vi. I contribute to the overall success of my organization, beyond just completing my job assignments.
vii. I feel comfortable suggesting alternative work plans or practices to my supervisor.
viii. I have voiced concerns to management about company policy without a negative consequence.
ix. I have received words of approval from my supervisor after suggesting a change in a particular work procedure, process, or plan

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Connections to OSH

An article in *Safety+Health* (Vargas, 2021) introduced psychological safety as a "hot concept in management" (p.42), and reported related interview statements from esteemed safety professionals. For example, Rosa Antonia Carrillo stressed the need for employees to "feel it's psychologically safe to give you bad news, to share mistakes, to share concerns or even give you critical feedback, because we can't improve without that feedback" (p. 43). Relatedly, Mark Hambrick, regional health and safety manager at Silent-Aire USA, reported that "psychological safety is the ability to have tough conversations and grow from them as an individual, group, team and organization" (p. 44).

As indicated earlier, the injury-prevention success of BBS depends on the delivery and acceptance of one-to-one behavioral feedback. After completing an employee-derived behavioral checklist, including reports of environmental factors that might have influenced the observed behavior, the observer should communicate the results to the observed worker. However, many organizations and their employees bypass this BBS process, and deliver the checklist information to an individual who enters the data on a computer program that computes averages for comparing safe and at-risk percentages across work teams, both within and between organizations.

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Why do many BBS applications omit the critical interpersonal feedback process? An obvious answer: Psychological safety is lacking. What does this mean in practical operational terms? What is lacking in an organization that omits the invaluable interpersonal conversation component of BBS? Yes, we are back to considering the safety culture—the environmental and social context of the workplace that can facilitate or stifle psychological safety.

Cultivating Psychological Safety

Given the obvious benefits of psychological safety, we arrive at the pivotal question: How can an organization cultivate a work culture that promotes, supports, and sustains psychological safety among its employees? A one-word answer to this vital and provocative question is simply “leadership.”

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The leaders of an organization set the cultural tone of that organization, thereby enabling or disenabling interpersonal trust and the frank and open dialogue that reflects a particular level of psychological safety.

Many safety professionals have authored articles that identify a variety of leadership characteristics that inspire and support psychological safety, including the author of this article (Geller, 2000, 2008, 2016). Here we consider six basic leadership qualities that have been empirically verified to distinguish the leaders of “good” versus “great” companies, and note their connection to encouraging and nurturing psychological safety. Many readers will recognize this reference to the classic best seller: “From Good to Great” by Jim Collins (2001).

More than two decades ago, Jim Collins and his research team studied eleven companies that transitioned from being good to being great, and they had sustained this level of excellence for at least 15 years. They systematically compared these good-to-great companies with a carefully-selected set of eleven companies that maintained good productivity and profits for at least 15 years, but never made the leap from good to great.

Collins and his research team found a number of common qualities among the good-to-great companies. Among those distinctions were the following leadership characteristics observed at the eleven good-to-great organizations.

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Please consider how each of these six leadership qualities relate to cultivating a work culture that facilitates psychological safety.

1. Manifest Personal Humility or Compelling Modesty

This leadership quality receives priority attention in Collins' book. He relates several case studies of companies that did not reach their potential because their leaders were more concerned about their own notoriety than the reputation of their company. The "good-to-great leaders never wanted to become larger-than-life heroes," rather they "were seemingly ordinary people quietly producing extraordinary results" (p. 28).

2. Project Success Beyond Self

Related to the first leadership quality, good-to-great leaders attribute company success to factors other than themselves. As systems thinkers, they see the big picture and realize their success is contingent on the daily small-win accomplishments of several individuals. They routinely acknowledge the synergistic contributions of the many employees who enable remarkable results.

3. Accept Responsibility for Failure

While spreading success beyond themselves, the good-to-great leaders take full responsibility for organizational failures. They face the brutal facts of less-than-desired outcomes, and they hold themselves accountable without blaming other people or just "bad luck." Interestingly, Collins and his team found that the leaders of the comparison companies often blamed others for lackluster performance, while taking personal credit for extraordinary results. Social psychologists call this the "self-serving bias" (Mezulis, Abramson, Hyde, & Hankin, 2004).

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4. Promote a Learning Culture

Humble leaders are open to new information. They are always learning, with a fervent belief in never-ending improvement. The good-to-great leaders facilitate fact finding to learn and improve, not to find someone to blame. They lead with questions rather than answers, and they promote frank and open dialogue and debate. The result: Employees are not satisfied with the status quo, but are continuously engaged in finding ways to improve company performance. They are constantly alert to possibilities for process refinement—diligently searching for the best solutions to problems, and regularly submitting suggestions to fine-tune their operations.

5. Work to Achieve, not to Avoid Failure

Although they reveal and face brutal facts, the good-to-great leaders never waver in their resolve for greatness. Failure is not an option; it is not even considered. With an optimistic stance, these leaders focus on achieving exemplary success. They attend to their envisioned enterprise with fanatic consistency and a disciplined constancy of purpose.

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6. Encourage Self-Motivation for Meaningful Work

Self-motivation is key to safe and long-term productivity and is enhanced whenever perceptions of personal choice or autonomy, competence, and community or relatedness are heightened (Deci & Flaste, 1995; Deci & Ryan, 1995; Geller, 2016b). Moreover, people are self-motivated when their behaviors provide natural ongoing consequences that are rewarding.

When does behavior on the job become intrinsically rewarding and self-motivating? Answer: When people believe their work is meaningful. When does this happen? Answer: Sometimes the special value of the effort is obvious, as when people are engaged in activities that prevent unintentional injuries. However, even in these cases, it is critical to provide interpersonal appreciation, gratitude, and supportive feedback to reassure people that they are accomplishing meaningful work. Great leaders know how to promote and support self-motivation, and they do this often.

This sixth leadership quality is especially significant for OSH, because it defines a self-transcendent source of self-directed motivation (Maslow, 1971). Specifically, working to keep people injury free is obviously meaningful actively-caring-for-people (AC4P) behavior that fuels self-motivation. In this regard, Jim Collins ends his best seller with the following: “It is impossible to have a great life unless it is a meaningful life. And it is very difficult to have a meaningful life without meaningful work” (p. 210). Safety leaders do meaningful work and therefore have meaningful lives.

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Great Leaders and Psychological Safety

The obvious connections between the qualities of good-to-great leaders, defined more than two decades ago by Jim Collins and his research team, and the leadership qualities purportedly needed to inspire and support psychological safety are quite remarkable. Indeed, Susan Vargas (2021) concluded her introduction to psychological safety by listing five ways for *anyone* to cultivate psychological safety: 1) Ask questions; 2) Be humble; 3) Build relationships; 4) Value all perspectives; and 5) Foster autonomy.

While managers and supervisors are assigned their corporate positions, any employee can choose to be a leader by practicing the six qualities of good-to-great leaders reviewed here as key to nurturing psychological safety. Please note also that these leadership qualities are relevant beyond the workplace to educational facilities, community-service agencies, sports teams, and the home.

Conclusion

This brief article addressed an important human dynamic that is becoming rather prevalent in the corporate world these days, including OSH. While the concept of “psychological safety” might seem new to some readers, it actually appeared in the management literature at least 30 years ago. Plus, while the safety and productivity advantages of cultivating a work, educational, or home culture that facilitates and supports psychological safety are obvious and demand serious consideration, such a mission or vision is certainly not new.

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As explained herein, the need to cultivate a culture of interpersonal trust, interdependent collaboration, and continuous improvement has been a primary mission of OSH since the 1980s or perhaps earlier. Indeed, the leadership strategies proposed to enhance and support psychological safety are the same as those identified empirically more than 20 years ago as qualities practiced by the leaders of especially successful organizations.

A revelation that a particular human dynamic is not new nor innovative does not detract from its relevance and importance. If a new label for a useful concept attracts more attention and implementation, then positive progress has been accomplished.

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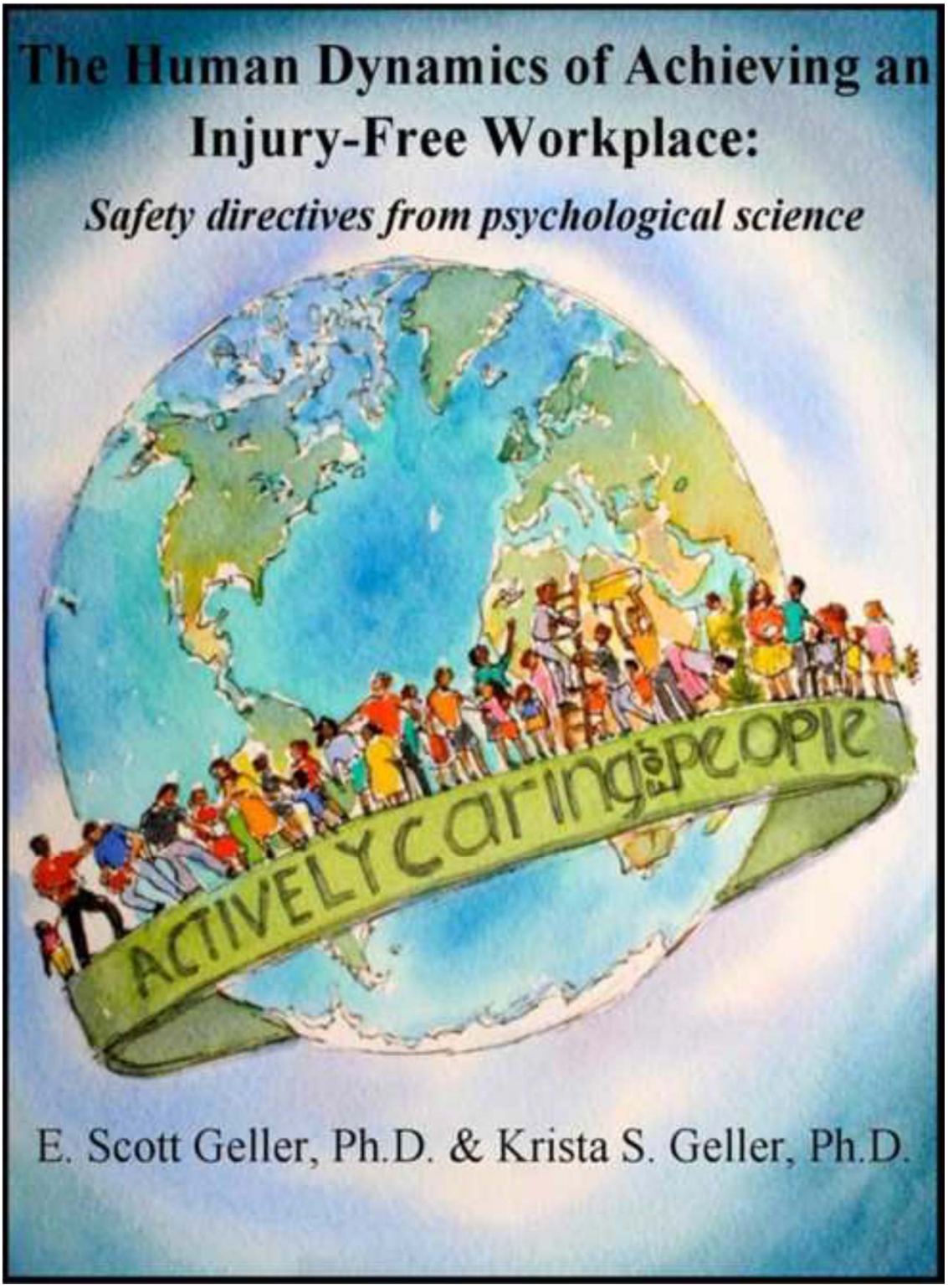
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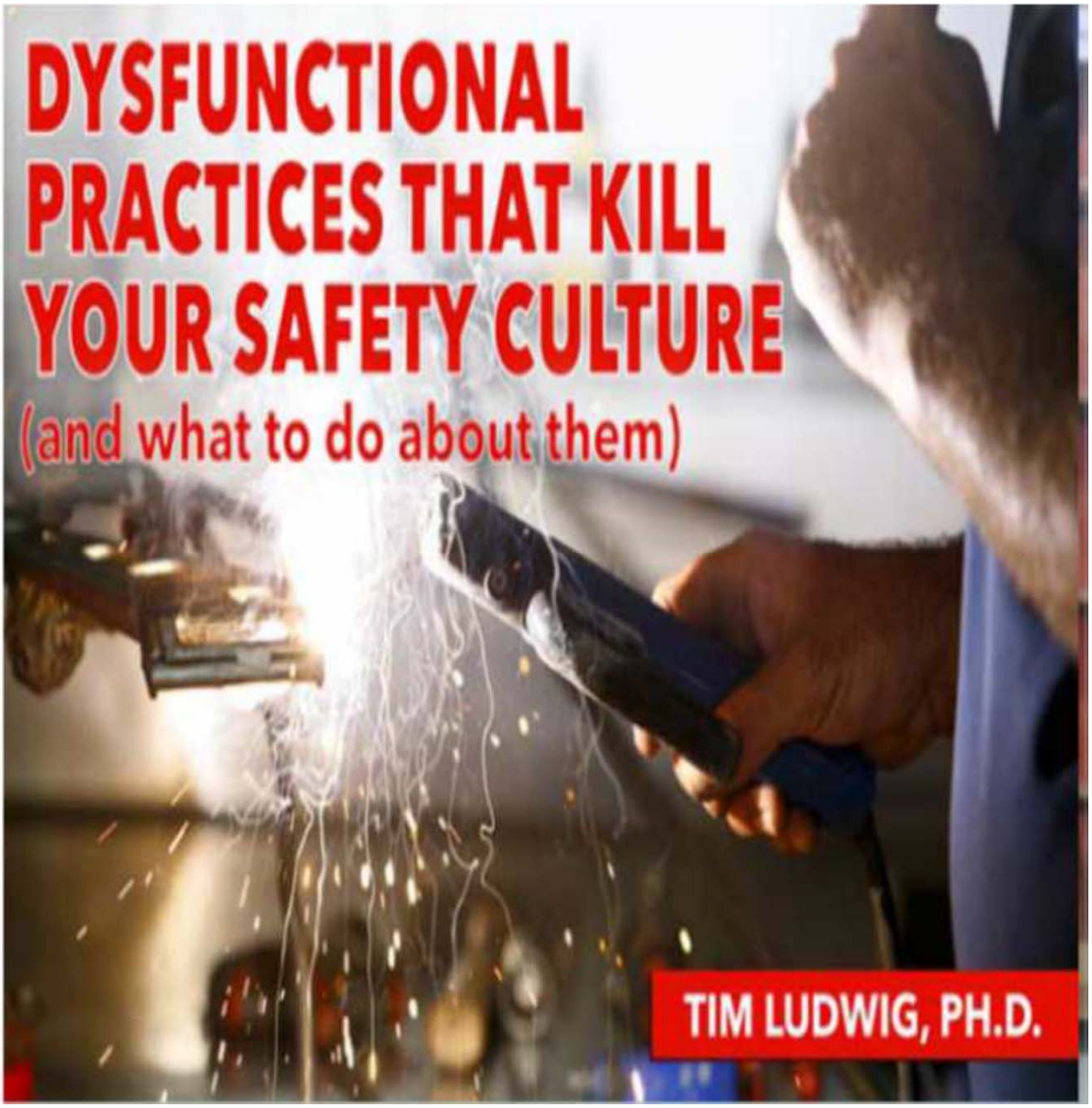
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Understanding the **"Dysfunctional Practices that kill your safety culture"** and learning what we can do about them is one step to breaking the safety plateau and achieving zero injury. The SHE Culture editorial team highly recommends this book to all EHS practitioners and industries seeking to achieve an injury free workplace – **SHE Culture Team**

"Labelling, blaming, fishing for faults and rushing to judge are among several dysfunctional practices that kill your safety culture. It is vital that we pay attention to them and understand what we can do about them." Labeling is a natural human tendency. "Stupid" is a label, no one is stupid. Labeling starts arguments and entrenches people against each other. You can't fix a label…." **- Dr. Tim Ludwig**

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Why Think Human Factor Engineering and Safety Leadership ?
Why Think Robotics & Digital Solutions for EHS problem solving ?
What is the future of Behavioral Safety as a tool for Loss Prevention ?

'Human Factors refer to environmental, organizational and job factors, and human and individual characteristics, which influence behaviour at work in a way which can affect health and safety. A simple way to view Human Factors is to think about three aspects: the job, the individual and the organization and how they impact on people's health and safety-related behaviour.' - ***UK HSE taken from its publication Reducing error and influencing behaviour (HSG48)***

'Human Factors is concerned with what people are being asked to do (the task and its characteristics), who is doing it (the individual and their competence) and where they are working (the organization and its attributes), all of which are influenced by the wider societal concern, both local and national.' ***– UK HSE Website***

Identifying the critical human factor concerns in every project and incorporating digital solutions driven by robotics and the integration of behavioral safety programmes will shape the future for loss prevention across various high risk industries be it process, aviation, construction etc.. ***Prof. Patricks-E, Chinemerem, EMT knowledge exchange forum 2024 edition***

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Human Factors Engineering & Human Error Precursors

Research has demonstrated that presence of various factors in combination increase the potential for error. These factors may be referred to as error precursors. Anticipation and identification of such precursors is a distinguishing performance strategy of highly performing individuals and organizations **(US DOE, 2012)**

The nature of the workplace, the task, personnel, and organizational issues are key factors with significant effect on human performance – **(The Energy Institute, 2008)**

Performance shaping factors also known as “preconditions”, or Performance Influencing Factors (PIFs) are the characteristics of the job, the individual, and the organization that influence human performance. Addressing these factors reduces the likelihood of all types of human failure. **(IOGP Report 621, 2018)**

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Human performance shaping factors and injury frequency rates in selected petroleum companies in Port Harcourt

Editor's Note & Commentary:
In this paper, the authors provided useful insights on the preconditions or key factors that influence Human performance through a study conducted across companies in Port Harcourt, Nigeria.

Disclaimer: *The data and facts presented here were not subjected to any verification by the SHE Culture review team and as such are purely research based findings by the authors within their study area.*

ABSTRACT
This study evaluated the impact of human performance shaping factors on injury frequency rates in selected petroleum companies in Port Harcourt. A cross-sectional research design was adopted for this study while purposive sampling technique was employed to select population for data collection. The population for the study was taken from four petroleum companies located in Port Harcourt. The determined sample size was one hundred and thirty-four (134). The data retrieved were analyzed using descriptive analysis, correlation, simple linear regression and analysis of variance (ANOVA). The results showed that job factors such as lack of planning and inadequate preparation for task, organizational factors such as poor organizational safety climate and safety culture and personal factors like fatigue and stress as well as lack of knowledge and training due to poor experience, inadequate workplace orientation and work process training were identified as key performance shaping factors responsible for most incidents and injuries. The results also revealed that there is a positive relationship between human performance shaping factors (job, organizational and person) and injury frequency rates with correlation coefficient value of 0.540, 0.689 and 0.583 respectively. Similarly, simple regression analysis with a value of 0.000 revealed that there is a significant impact of human performance shaping factors on incident rates. The same is true regarding the relationship between employee behaviour and human performance shaping factors given a p-value of 0.000 less than the alpha value (0.05). ANOVA results with F-statistic value of 1.302 and p-value of 0.277 revealed that there was no significant variation. In conclusion, the study confirmed that human performance shaping factors have an impact on injury frequency rates. It is therefore recommended that companies should adopt and implement behaviour-based safety program that modify human behaviour and improve safety performance. Also, human performance shaping factors should be considered while adopting approaches and resources for accident and incident prevention.
Key words: Human Performance Shaping Factors, Injury Prevention, Injury Frequency Rates

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Introduction

In recent years, there have been emphasis on incident prevention and safety management within the industry due to the need to protect human lives, environment and ensure business continuity. Some major causes of these incidents are linked to human factors. Studies have shown that over 80% of accidents are caused by human errors (Energy Institute, 2008). These factors encompass a range of elements related to human behaviour, decision-making and cognition, and they often interact with operational and technical aspects leading to failures and accidents (Energy Institute, 2008).

The causes of most of the notable industry incidents and injuries can be linked to human factors. For example, the Piper Alpha Disaster (1988) that killed over 167 people, the Texas City Refinery Explosion (2005) that resulted to deaths of about 15 people, the Deep-water Horizon rig disaster(2010) and the Buncefield Oil Depot Fire (2005) were attributed to human factor (Energy Institute, 2008). Despite the technological advancements, the role of human factors in incident prevention remains a crucial aspect to address. According to James Reason (1995), automation and increasingly advanced equipment do not cure human factors problems, they merely relocate them. Preventing these incidents is priority for the petroleum industries and understanding the impact of human performance shaping factors is essential to achieving this aim.

Human performance shaping factors (HPSFs) are various elements that influences and shape the behaviours, actions and decisions of individuals in a complex working environment such as the petroleum industry (HSE,1999). These factors encompass a wide range of aspects such as organizational culture, fatigue management, training, human-machine interface, workload, communication, leadership, work environment, motivational plan, organizational policies and more (HSE, 1999).

Understanding how these human performance shaping factors interact is essential for designing and implementing effective safety strategies for preventing incidents. According to Reason (1995), people do not act in isolation. Their behaviour is shaped by circumstances. The design of equipment, work processes and safety procedures must take into account human cognitive limitations, perceptions and response capabilities.

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The petroleum industry plays a vital role in the nation's economy, but it is inherently hazardous with the potential for serious accidents and injuries. Despite industry-wide safety measures and regulations, there exists a persistent and concerning trend of accidents and injuries within the industry. The frequent occurrences of accidents and injuries poses significant challenges to both the safety of workers and the operational efficiency of the petroleum companies. According to IOGP report (2021), the number of fatalities has increased from 14 in 2020 to 20 in 2021, the 20 fatalities occurred in 15 separate incidents. The overall total recordable injury rate (fatalities, lost workday cases, restricted workday cases and medical treatment cases) was 0.77, 10% higher than in 2020 (0.70) and in 2022, the fatal rate was 71% higher than in 2021 and the total recordable injury rate was 17% higher than in 2021.

Statistics show that most Nigerian petroleum companies have experienced several increase in First aid case, Medical Treated case, Restricted work case, near-misses and even fatality. These are evident in the Learning From Incidents (LFI) and safety alerts been shared across the industry. The study area, Port Harcourt is a hub for petroleum industry activities in Nigeria. Port Harcourt is the capital of Rivers State in southern Nigeria and is strategically located within the Niger Delta region. The Niger Delta is one of the world's largest and most prolific oil-producing regions, making Port Harcourt a key area for oil industry operators. The region hosts numerous oil and gas companies, both multinational and indigenous, making it an ideal study area. It has witnessed its share of industry related incidents. Most of the causes of these incidents can be linked to various human performance shaping factors. Hence, to address this issue effectively, it is imperative to assess the impact of human performance shaping factors on injury frequency rates within selected petroleum companies in Port Harcourt.

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It is perceived that performance shaping factors have the potential to influence worker behaviour and safety outcomes significantly. Therefore, understanding how these factors interact and contribute to injury frequency rates is crucial in order to plan effective preventive measures and enhance the overall safety culture in these organizations. Thus, the aim of the study was to provide valuable insights on the effects of human performance shaping factors on injury frequency rates in selected petroleum companies in Port Harcourt.

The study objectives were to:

i. Identify the human performance shaping factors responsible for most incidents and injuries in selected petroleum companies in Port Harcourt.
ii. Determine the relationship between human performance shaping factors and injury frequency rates in the selected petroleum companies in Port Harcourt.
iii. Examine the effects of human performance shaping factors on incident rates in selected petroleum companies in Port Harcourt.
iv. Assess the impact of human performance shaping factors on employee behaviour in selected petroleum companies in Port Harcourt.
v. Determine the variations of human performance shaping factors amongst selected petroleum companies in Port Harcourt.

Methodology

The study adopted the cross-sectional research design and was carried out in Port Harcourt, Rivers State, Nigeria. The population for the study comprised four (4) selected petroleum companies within Port Harcourt, Nigeria which are identified as A, B, C, and D for the purpose of confidentiality. All the companies carry out their operations onshore. The study did not consider petrol station workers, their transport personnel and tank farm staff. The selection of these companies was based on their characteristics or feature as petroleum industry operating upstream, midstream, or downstream in Nigeria.

The survey questionnaire and checklist were the two research study instruments deployed for this research. A total of one hundred and forty-nine (149) copies of the survey questionnaire and checklists were administered to participants following a purposive sampling technique. The distribution was as follows: thirty (30) copies were administered to company A, forty-four (44) copies were administered to company B, twenty-four (24) copies were administered to company C, and fifty-one (51) copies were administered to company D. A total of one hundred and thirty-nine (139) copies out of the one hundred and forty-nine (149) administered questionnaire were returned and used for data analysis.

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Descriptive and inferential statistics were used for the data analysis. Correlation was used to analyze the relationship between performance shaping factors and injury frequency rate. Simple regression was used to analyze the effects and impacts of HPSF's on employee behaviour and incident rates as well as to analyze the effects and impact of performance shaping factors on employee behaviour and injury frequency rate. ANOVA was used to determine the variations across the selected petroleum companies.

Results and Discussion

Key human performance shaping factors responsible for most incidents and injuries

With respect to Job factors, respondents agreed that lack of planning and inadequate preparation for task such as permits, risk assessments, pre-start checks contributed to incidents. Table 1: shows the key HPSF responsible for most incidents and injuries. These job factors had a greater mean value compared to other responses and a lower standard deviation which indicated that there were stability of idea on the specific variable.

This result is in consonance with a study by Smith et al. (2018) which emphasized that inadequate planning significantly increases the likelihood of accidents, with a direct correlation between poor preparation and incident rates. The study supports that insufficient attention to tasks such as obtaining permits, conducting thorough risk assessments, and completing pre-start checks can create a breeding ground for potential disasters. Also, research conducted by Johnson and Brown (2017) which identified the absence of proper planning as a major factor in accidents related to construction support this finding; the study noted that overlooking essential steps like obtaining necessary permits contributes to a higher occurrence of on-site incidents. Additionally, a comprehensive review by Safety Dynamics (2020) which delved into the consequences of inadequate preparation in the oil and gas industry highlighted that failure to meticulously plan tasks such as well-site operations and emergency response procedures significantly heightens the potential for severe incidents and injuries.

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Table 1: Identified HPSF Related to Job Factors

Items	SA (5)	A (4)	UD (3)	D (2)	SD (1)	Weighted Mean	Rank	Decision
The lack of clarity and absence of safety signs, signals, instructions, and other information have contributed to visible causal factors to incidents	98(490)	35(140)	2(6)	2(4)	2(2)	4.62	4th	Agree
Most safety alerts and incident notification identified difficulty/complexity of task as one of the causes of injuries	21(105)	62(248)	16(48)	34(68)	6(6)	3.42	8th	Agree
Work overload is a causal factor to incidents and injuries	61(305)	57(228)	4(12)	15(30)	2(2)	4.15	7th	Agree
The absence of safe work procedures or the inadequacy or inappropriateness of safe work procedures can be a major cause of incidents and injuries	98(490)	41(164)	0(0)	0(0)	0(0)	4.71	3rd	Agree
Lack of planning and inadequate preparation for task such as permits, risk assessments, pre-start checks contribute to incidents	110(550)	29(116)	0(0)	0(0)	0(0)	4.79	1st	Agree
Unsafe working environment such as high noise, high temperatures/heat, restricted or confined space, poor lighting and ventilation can cause incidents and injuries	107(535)	28(112)	4(12)	0(0)	0(0)	4.74	2nd	Agree
Poor communication with colleagues, supervision, and contractor contributes to injuries and incidents	77(385)	57(228)	1(3)	(0)	0(0)	4.49	6th	Agree
Tools appropriate for task can help prevent incidents	85(425)	48(192)	0(0)	6(12)	0(0)	4.53	5th	Agree
Total	657(3285)	357(1428)	27(81)	61(122)	10(10)	4.43		Agree

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Results from Table 2 also revealed that with consideration to Organizational factors, respondents agreed that poor organizational safety climate and safety culture are among the causes of injuries and incidents. This result agrees with the research by James Reason which stated that Organizational culture plays a significant role in shaping employee behaviour. A positive safety culture encourages reporting of errors and near-misses, while a punitive culture can deter reporting (Reason, 1997). Also, Research by Schein (1990) which emphasizes that culture shapes behaviour and decision-making within an organization as positive safety culture encourages open communication, reporting, and learning from errors, while a blame-oriented culture can suppress reporting and hinder improvement efforts (Hudson et al., 2003) further supported the findings.

Table 2: Identified HPSF Related to Organization Factors

Items (n=139)	SA (5)	A (4)	UD (3)	D (2)	SD (1)	Weighted Mean	Rank	Decision
Work pressures created by ambitious attention and focus on meeting production targets, profit at the expense of safety contributes to incidents and injuries	64(320)	65(260)	6(18)	4(8)	0(0)	4.36	2nd	Agree
Poor organizational Safety Climate and Safety culture are among the causes of injuries and incidents	87(435)	46(184)	6(18)	0(0)	0(0)	4.58	1st	Agree
Negative Peer pressure and influence can cause incidents in most petroleum companies	35(175)	57(228)	28(84)	17(34)	2(2)	3.76	5th	Agree
Lack of clarity of roles and responsibilities have contributed to most incidents	59(295)	67(268)	11(33)	2(4)	0(0)	4.32	4th	Agree
Poor Supervision or absence of supervision and safety leadership is among the causes of incidents and injuries	64(320)	66(264)	4(12)	5(10)	0(0)	4.36	2nd	Agree
Inadequate number of people engaged in or required for work is the cause of most incidents	34(170)	62(248)	18(54)	23(46)	2(2)	3.74	6th	Agree
Total	343(1715)	363(1452)	73(219)	51(102)	4(4)	4.19		Agree

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Results from Table 3 focused on person factors which revealed that respondents agreed that fatigue and stress can constitute to incidents and injuries. They also agreed that lack of knowledge and poor experience due to inadequate training, inadequate workplace orientation and work process training is a causal factor to incidents and injuries. This finding is in conformity with the study conducted by Lamound (2008) which revealed that fatigue is a well-documented factor that can impair cognitive and physical performance and that Sleep deprivation and long working hours have been associated with increased incidents. Also, study by Hobfoll (2002) showed that high levels of stress and anxiety can impair decision-making and situational awareness. Stressors in the workplace, such as time pressure, can lead to errors. Similarly, the study from Cannon-Bowers and Salas (2015) supported the findings. Their study revealed that the quality and effectiveness of training programs can influence human performance and that well-trained individuals are more likely to make better decisions in high-pressure situations. Adequate training and competency assessment are crucial to ensuring that individuals have the necessary skills and knowledge to perform their tasks safely. Also, studies by Salas et al. (2008) stressed the importance of ongoing training, simulation, and competency assessment in high-risk industries.

Table 3: Identified HPSF Related to Person Factors

Items	SA (5)	A (4)	UD (3)	D (2)	SD (1)	Weighted Mean	Rank	Decision
Physical capability and condition are factors that contributes to most incidents and accidents	33(165)	63(252)	19(57)	20(40)	4(4)	3.73	6th	Agree
Fatigue and stress can contribute to incidents and injuries	98(490)	39(156)	0(0)	2(4)	0(0)	4.68	1st	Agree
Divided attention was identified as one of the causes of injuries	84(420)	53(212)	2(6)	0(0)	0(0)	4.59	3rd	Agree
Lack of knowledge and training due to poor experience, inadequate workplace orientation and work process training is a causal factor to incidents and injuries	95(475)	44(176)	0(0)	0(0)	0(0)	4.68	1st	Agree
Lack of Competence to deal with circumstances and poor situational awareness contributes to incidents	77(385)	60(240)	0(0)	0(0)	2(2)	4.51	4th	Agree
Lack of motivation and low morale cause accidents	42(210)	73(292)	17(51)	5(10)	2(2)	4.06	5th	Agree
Total	429(2145)	332(1328)	38(114)	27(54)	8(8)	4.38		Agree

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Relationship between human performance shaping factors and injury frequency rates

Results from Table 4 revealed that respondents agreed that there is a significant relationship between human performance shaping factors and injury frequency rates in the selected petroleum companies in Port Harcourt. Spearman's rank correlation reveals that job factor, organizational factor and person factor with ($p<0.05$) correlates injury frequency rate with a positive correlation coefficient value of 0.540, 0.689 and 0.583 respectively (Table 5). Partial correlation (Table 6) further revealed that employee behaviour with ($p<0.05$) has a significant relationship between human performance shaping factors and injury frequency rates with a positive correlation of 0.392, 0.613 and 0.431 respectively. Simple regression revealed that injury frequency rate (dependent variable) was significantly impacted by the independent variable (human performance shaping factors) with p-value (0.000) < 0.05. More so, the coefficient of determination (R2) (Table 7) showed that 54% of injury frequency rate is determined by human performance shaping factors. The model equation (Table 8) showed that the coefficient of human performance shaping factors positively impacts injury frequency rate. Thus, increase in human performance shaping factors without proper management has a corresponding increase in injury frequency rate in selected petroleum companies in Port Harcourt.

These findings corroborates with James Reason's Swiss Cheese Model of accident causation theory which suggests that multiple layers of defenses (represented as slices of Swiss cheese) can be penetrated by active failures (e.g., human errors) and latent conditions (e.g., HPSFs). These latent conditions, including factors like fatigue, complacency, and stress, can increase the likelihood of human errors, thereby affecting injury rates (Reason, 2000). Similarly, research carried out by Kirwan (1994) on the Human Error Assessment and Reduction Technique (HEART) is consistent with the finding. The study highlighted on HEART which is a widely used method to assess the impact of HPSFs on error probability as a model which incorporates factors such as task complexity, workload, and situational awareness and provided insights into how these factors influence the likelihood of errors and, consequently, injuries. Also, the research by Weick and Sutcliffe (2007) further agrees with the findings. The research showed that organizations with a strong safety culture and effective management of HPSFs tend to have lower injury frequency rates. Furthermore, studies by Zohar (2010) have also shown that organizations with a strong safety culture, characterized by open communication, shared values, and a commitment to safety, tend to have lower injury rates which is in consonance with the findings from the reports. Similarly, study by Clarke (2010) which asserts that insufficient training and low skill levels are associated with higher injury rates supports the findings. Correspondingly, study by Carayon (2006) which assert that high workload and fatigue, often influenced by job/task factors, are linked to an increased risk of injuries further supports the findings.

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Table 4: Respondents Opinion on Injury Frequency Rates HPSF

Items	SA (5)	A (4)	UD (3)	D (2)	SD (1)	Weighted Mean	Rank	Decision
Most first aid cases have been associated with fatigue and stress	23(115)	71(284)	2(6)	19(38)	0(0)	3.71	10th	Agree
Lessons learned from incidents (LFI) shows that Lost workday cases can be linked to lack of experience and incompetence of personnel	27(135)	70(280)	29(87)	11(22)	2(2)	3.78	9th	Agree
Medical treatment cases have risen following the absence of supervision and safe systems of work	31(155)	77(308)	18(54)	9(18)	4(4)	3.88	8th	Agree
Safety alerts have identified Near misses to have been linked to poor workplace condition and poor safety culture	54(270)	68(272)	7(21)	8(16)	2(2)	4.18	6th	Agree
Horse playing in the form of distracting colleagues at work, teasing, abusing, startling, quarrelling, practical joking, throwing material and tools, showing off, shouting and making disturbances have cased high potential near misses and fatal accidents	68(340)	58(232)	5(15)	4(8)	4(4)	4.31	4th	Agree
Improper use of tools and equipment have led to personnel injury that resulted to both restricted work cases and lost work cases	77(385)	59(236)	1(3)	2(4)	0(0)	4.52	2nd	Agree
The wrong use of hands and other parts of the body have resulted to medical treatment cases and ergonomic injuries	53(265)	64(256)	17(51)	5(10)	0(0)	4.19	5th	Agree
Lost time injuries have been linked to task demands, conditions of the working environment and individual capabilities or person factors	23(115)	94(376)	18(54)	2(4)	2(2)	3.96	7th	Agree
Most fatalities and lost time injuries were caused by the lack of safe working tools, poor planning and absence of task specific safe systems of work such as JHA, risk Assessment and PTW	78(390)	56(224)	2(6)	3(6)	0(0)	4.50	3rd	Agree
Using defective tools and other uncertified equipment for work such as lifting slings, scaffolds, ladders, that are in poor condition have created opportunities for damage and personnel injury including fatality	101(505)	36(144)	2(6)	0(0)	0(0)	4.71	1st	Agree
Total	535(2675)	653(2612)	125(375)	63(126)	14(14)	4.17		Agree

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Table 5: Correlations on Injury Frequency Rates Statements

			JF	OF	PF	INFT
Spearman's rho	JF	Correlation Coefficient	1.000	.564**	.477**	.540**
		Sig. (2-tailed)	.	.000	.000	.000
		N	139	139	139	139
	OF	Correlation Coefficient	.564**	1.000	.577**	.689**
		Sig. (2-tailed)	.000	.	.000	.000
		N	139	139	139	139
	PF	Correlation Coefficient	.477**	.577**	1.000	.583**
		Sig. (2-tailed)	.000	.000	.	.000
		N	139	139	139	139
	INFT	Correlation Coefficient	.540**	.689**	.583**	1.000
		Sig. (2-tailed)	.000	.000	.000	.
		N	139	139	139	139

**. Correlation is significant at the 0.01 level (2-tailed).

Table 6: Partial Correlations on Injury Frequency Rates Statements

Control Variables			JF	OF	PF	INFT
EB	JF	Correlation	1.000	.365	.309	.392
		Significance (2-tailed)	.	.000	.000	.000
		df	0	136	136	136
	OF	Correlation	.365	1.000	.487	.613
		Significance (2-tailed)	.000	.	.000	.000
		df	136	0	136	136
	PF	Correlation	.309	.487	1.000	.431
		Significance (2-tailed)	.000	.000	.	.000
		df	136	136	0	136
	INFT	Correlation	.392	.613	.431	1.000
		Significance (2-tailed)	.000	.000	.000	.
		df	136	136	136	0

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Table 7: Statistical Summary on Injury Frequency Rates Statements

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics					Durbin-Watson
					R Square Change	F Change	df1	df2	Sig. F Change	
1	.734[a]	.539	.528	3.152	.539	52.528	3	135	.000	2.155

a. Predictors: (Constant), PF, JF, OF b. Dependent Variable:INFT

Table 8: Coefficients on Injury Frequency Rates Statements

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	3.024	3.334		.907	.366
	JF	.344	.094	.249	3.648	.000
	OF	.666	.112	.434	5.936	.000
	PF	.373	.132	.207	2.835	.005

a. Dependent Variable: INFT

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Impact of human performance shaping factors on incident rates

Results from Table 9 reveals that job factor, organizational factor and person factor significantly with ($p<0.05$) correlates incident rate with a positive correlation coefficient value of 0.524, 0.560 and 0.540 respectively (Table 10). The coefficient of determination (R2) (Table 12) showed that 40% of incident rates are determined by human performance shaping factors. The model equation showed that the coefficient of human performance shaping factors positively affects incident rate (Table 13). Thus, increase in human performance shaping factors has a corresponding increase in incident rates in selected petroleum companies in Port Harcourt. Therefore, the relationship between human performance shaping factors and incident rates in selected petroleum companies in Port Harcourt is significant. This finding is in consonance with Heinrich's work; Heinrich's Domino Theory (1931) suggests that the root cause of most workplace accidents is the unsafe acts of people, which can be influenced by factors such as worker attitudes and behaviours. Again, James Reason's Swiss cheese model emphasized that incidents occur when multiple barriers (representing human and system factors) align, allowing errors to pass through. It highlights the role of human factors in incident causation. Similarly, Hudson's Human Factors Analysis and Classification System (HFACS, 2003) also identified various human performance shaping factors, including organizational and supervisory influences, which can contribute to incidents in aviation and other high-risk industries, and this is consistent with the findings.

In addition, Erik Hollnagel's work on Resilience Engineering (2006) acknowledged that humans are an essential part of complex systems and that understanding their performance shaping factors is crucial for preventing incidents; this is consistent with the findings. Similarly, Salas et al., (2015) in the work "The Five Pillars of High-Reliability Organizations" discussed the importance of human performance shaping factors in high-reliability organizations and how they contribute to safety and incident prevention, this also corroborates the findings from the study.

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Table 9: Respondents opinion on Impact of HPSF's on Incident Rates

Items	SA (5)	A (4)	UD (3)	D (2)	SD (1)	Weighted Mean	Rank	Decision
Improving knowledge, skill and competence of workers through training and awareness creation will better the workplace conditions and eliminate human and operating errors that causes accidents	100(500)	32(128)	3(9)	2(4)	2(2)	4.63	3rd	Agree
Improving safety leadership visibility and supervision to reward and recognize safe behaviours will eliminate complacency, horse play, short-cuts, failure to wear PPE and related human errors and prevent injuries	97(485)	38(152)	0(0)	4(8)	0(0)	4.64	2nd	Agree
Providing adequate Standards, policies, procedures (PSP) will enable workers to use correct tools, avoid taking short cuts, ensure adequate and timely warning and avoid unauthorized deviations from safe work systems that can prevent incidents and injuries	95(475)	32(128)	2(6)	8(16)	2(2)	4.51	7th	Agree
Improvement of workplace assets through equipment maintenance schedules will reduce equipment wear and tear and use of defective tools and equipment by workers and create an injury free workplace	86(430)	46(184)	3(9)	4(8)	0(0)	4.54	6th	Agree
Improving Proper communication will ensure respect and compliance to safe systems of work and safe work procedures and eliminate the risk of injuries	89(445)	47(188)	1(3)	2(4)	0(0)	4.60	4th	Agree
Better procurement and material supplies will eliminate frequent use of unsafe and defective equipment and the potential incidents and injuries	78(390)	55(220)	2(6)	4(8)	0(0)	4.49	8th	Agree
Management commitment to safety culture, awareness and accountability can significantly reduce incident rates	9(45)	48(192)	0(0)	0(0)	0(0)	4.65	1st	Agree
Reward/recognition system can improve workers morale and encourage them to carry out tasks safely, hence reducing potential incidents	76(380)	57(228)	4(12)	2(4)	0(0)	4.49	8th	Agree
Consequence management for failure to follow rules/procedures can aid in incident prevention	51(255)	63(252)	9(27)	12(24)	4(4)	4.04	10th	Agree
Good working environment such as appropriate noise level, heat and space, lighting and ventilation system can enable better working performance from workers and reduce the likelihood of incidents	81(405)	56(224)	2(6)	0(0)	0(0)	4.57	5th	Agree
Total	844(4220)	474(1896)	26(78)	38(76)	8(8)	4.52		Agree

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Table 10: Correlation on Impact of HPSF's On Incident Rates

			JF	OF	PF	INT
Spearman's rho	JF	Correlation Coefficient	1.000	.564**	.477**	.524**
		Sig. (2-tailed)	.	.000	.000	.000
		N	139	139	139	139
	OF	Correlation Coefficient	.564**	1.000	.577**	.560**
		Sig. (2-tailed)	.000	.	.000	.000
		N	139	139	139	139
	PF	Correlation Coefficient	.477**	.577**	1.000	.540**
		Sig. (2-tailed)	.000	.000	.	.000
		N	139	139	139	139
	INT	Correlation Coefficient	.524**	.560**	.540**	1.000
		Sig. (2-tailed)	.000	.000	.000	.
		N	139	139	139	139

**. Correlation is significant at the 0.01 level (2-tailed).

Table 11: Partial Correlation on Impact of HPSF's On Incident Rates

Control Variables			JF	OF	PF	INFT
EB	JF	Correlation	1.000	.365	.309	.392
		Significance (2-tailed)	.	.000	.000	.000
		Df	0	136	136	136
	OF	Correlation	.365	1.000	.487	.613
		Significance (2-tailed)	.000	.	.000	.000
		Df	136	0	136	136
	PF	Correlation	.309	.487	1.000	.431
		Significance (2-tailed)	.000	.000	.	.000
		Df	136	136	0	136
	INFT	Correlation	.392	.613	.431	1.000
		Significance (2-tailed)	.000	.000	.000	.
		Df	136	136	136	0

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Table 12: Statistical Summary on Impact of HPSF's On Incident Rates

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics					Durbin-Watson
					R Square Change	F Change	df1	df2	Sig. F Change	
1	.635[a]	.403	.390	3.604	.403	30.418	3	135	.000	1.724

a. Predictors: (Constant), PF, JF, OF b. Dependent Variable: INT

Table 13: Coefficient on Impact of HPSF's On Incident Rates

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	9.680	3.812		2.539	.012
	JF	.400	.108	.288	3.714	.000
	OF	.392	.128	.254	3.057	.003
	PF	.436	.151	.240	2.895	.004

a. Dependent Variable: INT

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Impact of human performance factors on employee behaviour

Table 14 revealed that respondents agree that human performance shaping factors has an influence on employee behaviour in the selected petroleum companies in Port Harcourt (r=0.539) (Table 15). This finding agrees with the research conducted by Barrick and Mount (1991) which showed that personality traits like conscientiousness and agreeableness can significantly affect an individual's behaviour in the workplace. Also, it corroborates with Herzberg's Two-Factor Theory (1959) which highlights the importance of motivation in employee behaviour. It stated that factors like job satisfaction and intrinsic motivation can lead to increased productivity and positive behaviours. Similarly, study by Salas which agrees that investing in employee training and development is a critical HPSF that can impact employee behaviour. In addition, the finding is consistent with reports from Eisenbeiss (2008) which highlights how reward systems influence employee's behaviour. The Study showed that well-designed reward and recognition programs can motivate employees to exhibit behaviours aligned with organizational goals. Furthermore, transformational leadership, as discussed by Bass and Riggio (2006), also supports the findings. The study suggested that transformational leadership can positively influence employee behaviour by inspiring and motivating them to go above and beyond their job roles. Correspondingly, study explored by Schein (2010) on Organizational culture, supports the finding. Study highlights that organizational culture plays a pivotal role in shaping employee behaviour. It agrees that a culture that promotes collaboration and innovation is likely to result in proactive and positive behaviours.

Table 14: Respondents Opinion on Impact of HPSF's On Employees' Behaviour

Items	SA (5)	A (4)	UD (3)	D (2)	SD (1)	Weighted Mean	Rank	Decision
Recognition and rewards for good performance motivates me to perform better	72(360)	45(180)	14(42)	8(16)	0(0)	4.30	4th	Agree
Adequate training and development opportunities are provided to enhance my performance	76(380)	51(204)	2(6)	6(12)	4(4)	4.36	2nd	Agree
Work-life balance is maintained in my organization, leading to positive employee behaviour	53(265)	64(256)	13(39)	7(14)	2(2)	4.14	7th	Agree
The clarity of job expectations in my organization positively influences my behaviour	51(255)	79(316)	5(15)	2(4)	2(2)	4.26	5th	Agree
Effective communication within my organization positively impacts my behaviour	62(310)	70(280)	1(3)	4(8)	2(2)	4.34	3rd	Agree
Leadership and management practices in my organization encourage employee performance	70(350)	57(228)	6(18)	5(10)	1(1)	4.37	1st	Agree
I have access to the necessary resources and tools to perform my job effectively	53(265)	66(264)	12(36)	8(16)	0(0)	4.18	6th	Agree
Total	437(2185)	432(1728)	53(159)	40(80)	11(11)	4.28		Agree

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Table 15: Model Summary on Impact of HPSF's On Employees' Behaviour

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics					Durbin-Watson
					R Square Change	F Change	df1	df2	Sig. F Change	
1	.539[a]	.291	.275	3.556	.291	18.457	3	135	.000	2.065

a. Predictors: (Constant), PF, JF, OF

b. Dependent Variable: EB

Table 16: Coefficient on Impact of HPSF's On Employees' Behaviour

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	2.224	3.761		.591	.555
	JF	.427	.106	.340	4.014	.000
	OF	.007	.127	.005	.055	.956
	PF	.473	.149	.288	3.185	.002

a. Dependent Variable: EB

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Variations of human performance shaping factors amongst selected petroleum companies in Port Harcourt

Results in Table 17 revealed that there were no variations regarding human performance shaping factors amongst selected companies in Port Harcourt. Results from Table 18 revealed that the various companies are not significantly different with F-statistic value of 1.302 and p-value of 0.277 (Table 18). This implies that there is no significant difference in variations of human performance shaping factors among the selected petroleum companies

Table 17: Respondents Opinion on Variations of Human Performance Shaping Factors

Companies	A (n=28)		B (n=24)		C (n=40)		D (n=47)		Total (n=139)	
Items	Y (%)	N (%)	Y (%)	N (%)	Y (%)	N (%)	Y (%)	N (%)	Y (%)	N (%)
Are the observed and witnessed incidents caused by cases due to task demands?	12(42.9)	16(57.1)	10(41.7)	14(58.3)	22(55)	18(45)	23(48.9)	24(51.1)	67(48)	72(52)
Were the observed medical treatment cases due to work environment issues?	11(39.3)	17(60.7)	8(33.3)	16(66.7)	18(45)	22(55)	33(70.2)	14(29.8)	70(50)	69(50)
Were the restricted work cases due to individual capability and lack of competence	18(64.3)	10(35.7)	12(50)	12(50)	26(65)	14(35)	23(48.9)	24(51.1)	79(57)	60(43)
Did human nature contribute to any of the incidents?	27(96.4)	1(3.6)	22(91.7)	2(8.3)	30(75)	10(25)	40(85.1)	7(14.9)	119(86)	20(14)
Were the lost time injuries (LTI) caused by line of fire errors and hazards?	23(82.1)	5(17.9)	16(66.7)	8(33.3)	28(70)	12(30)	37(78.7)	10(21.3)	104(75)	35(25)
Were the LTI caused by poor and awkward posture?	17(60.7)	11(39.3)	12(50)	12(50)	20(50)	20(50)	21(44.7)	26(55.3)	70(50)	69(50)
Were the LTI caused by tools and materials placed in wrong places and general housekeeping issues?	20(71.4)	8(28.6)	22(91.7)	2(8.3)	32(80)	8(20)	37(78.7)	10(21.3)	111(80)	28(20)
Were the LTI caused by observed cases of dust in air?	5(17.9)	23(82.1)	4(16.7)	20(83.3)	10(25)	30(75)	11(23.4)	36(76.6)	30(22)	109(78)
Were the observed incidents caused by strong wind effect and other poor weather conditions?	11(39.3)	17(60.7)	4(16.7)	20(83.3)	18(45)	22(55)	29(61.7)	18(38.3)	62(45)	77(55)
Were the incidents due to height and elevation issues?	10(35.7)	18(64.3)	10(41.7)	14(58.3)	26(65)	14(35)	30(63.8)	17(36.2)	76(55)	63(45)
Were the LTI caused by issues with restricted spaces and illumination?	15(53.6)	13(46.4)	12(50)	12(50)	20(50)	20(50)	27(57.4)	20(42.6)	74(53)	65(47)
Were the LTI caused by poor ventilation?	14(50)	14(50)	4(16.7)	20(83.3)	20(50)	20(50)	24(51.1)	23(48.9)	62(45)	77(55)
Were the LTI related to issues with 1ise?	11(39.3)	17(60.7)	6(25)	18(75)	18(45)	22(55)	24(51.1)	23(48.9)	59(42)	80(58)
Were the LTI related with any issues of temperature and heat?	13(46.4)	15(53.6)	8(33.3)	16(66.7)	18(45)	22(55)	23(48.9)	24(51.1)	62(45)	77(55)
Total	207 (52.8)	185 (47.2)	150 (44.6)	186 (55.4)	306 (54.6)	254 (45.4)	382 (58.1)	276 (41.9)	1045 (54)	901 (46)

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Table 18: ANOVA on human performance shaping factors

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	.291	3	.097	1.302	.277
Within Groups	10.069	135	.075		
Total	10.361	138			

Conclusion

A study on the effects of human performance shaping factors on injury frequency rates in selected petroleum companies in Port Harcourt was conducted; Job factors, such as lack of planning and inadequate preparation for task such as permits, risk assessments, pre-start check, organizational factors such as poor organizational safety climate and safety, person factors like fatigue and stress as well as lack of knowledge and training due to poor experience, inadequate workplace orientation and work process training were identified as the key performance shaping factors responsible for most incidents and injuries.

The study further revealed that there is a significant relationship between human performance shaping factors and injury frequency rates. Thus, increase in human performance shaping factors has a corresponding increase in injury frequency rate in selected petroleum companies in Port Harcourt. Also, simple regression analysis revealed that incident rate (dependent variable) was influenced by the independent variable (human performance shaping factors) with p-value (0.000) < 0.05 indicating a significant level of impact. Again, simple regression analysis further revealed that the relationship between employee behaviour (dependent variable) and the independent variable (human performance shaping factors) is significant since p-value (0.000) < 0.05. However, investigating each human performance shaping factors, only job factor and person factor significantly impact employee's behaviour. More so, the coefficient of determination (R2) showed that 29% of employees' behaviour is determined by human performance shaping factors. In addition, the results of ANOVA revealed that the various companies are not significantly different with F-statistic value of 1.302 and p-value of 0.277.

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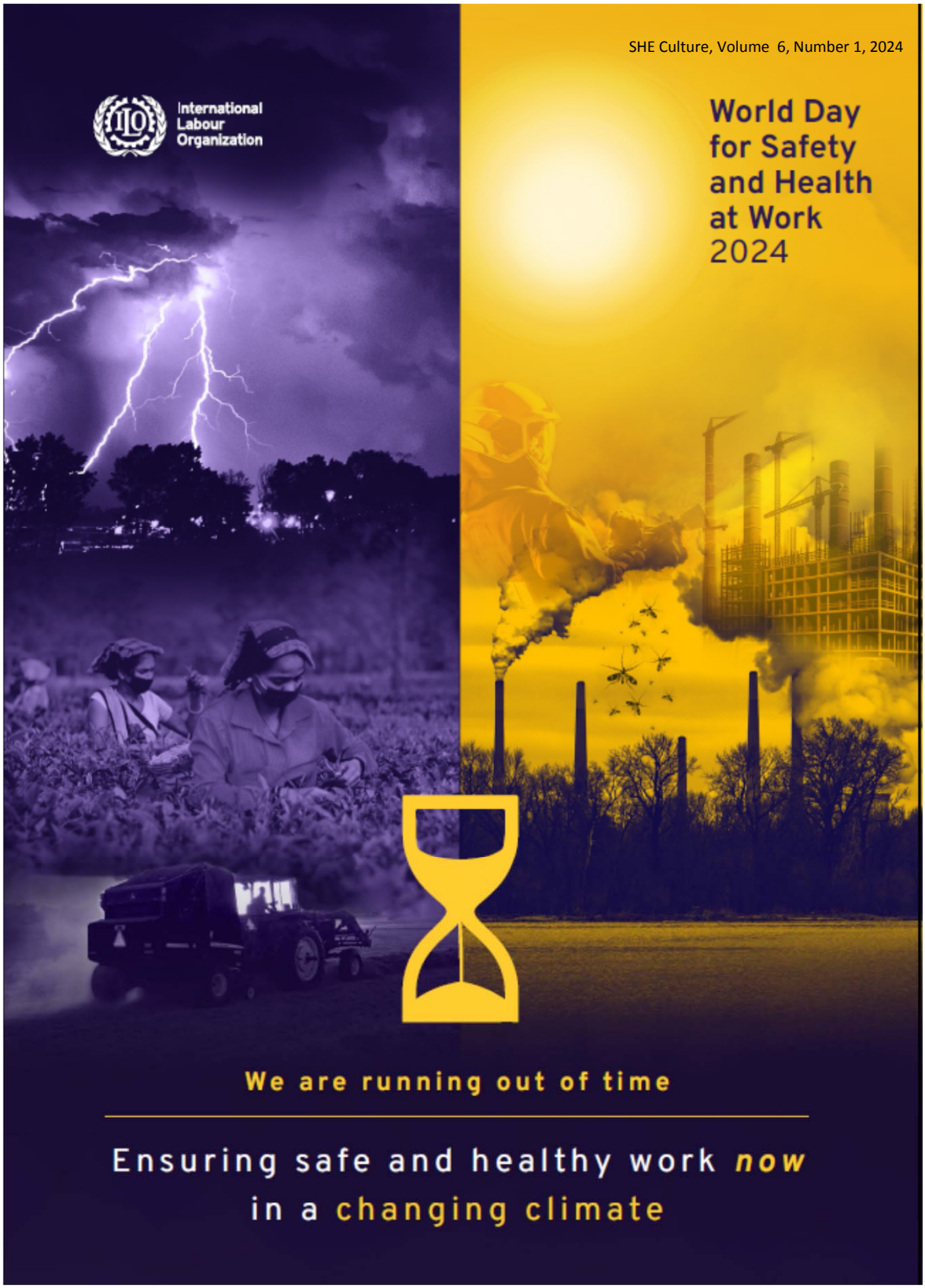
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"Every 28 April, the International Labour Organization commemorates the World Day for Safety and Health at Work, focusing on a timely theme related to occupational safety and health. This year, the theme will focus on exploring the impacts of climate change on occupational safety and health" – ***ILO website, 2024***

How would climate change impact on employee health & safety in the future work place? Have you ever given a thought to the effects of rising temperatures and Green House Gases (GHG)?

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Understanding Climate Change & Building Resilience: Focus on Africa — Dr. (Sir) Patricks-E. Chinemerem, FCAI

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Building Sustainable & Climate Resilient Future for Africa through adaptation, mitigation and climate change education



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